

Quadratic-logarithmic Gaussian dimension for all integer cell counts

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The spherical-cap converse proves that bounded $O(1/k)$ additive error to the unrestricted equal-mass Gaussian centroid optimum requires dimension $\Omega((\log k)^2)$ for every k . We prove the **matching sufficient dimensional order for every sufficiently large integer number of cells**. The first stage uses binary linear codes at dyadic cardinalities; a one-dimensional Gaussian selector and a bounded-entropy binary expansion then extend it to **all** integers k . Every partition block has exact Gaussian mass $1/k$.

The only substantial classical external probabilistic input is the Berry–Esseen inequality for independent, not necessarily identically distributed, centered summands. We state the exact form and constants used and trace every application below. All other probabilistic estimates are derived directly.

Main theorem

All-integer conclusion (Theorem 11). For every sufficiently large integer k , without arithmetic restrictions on k , the least Gaussian dimension needed for additive $92/k$ accuracy satisfies

$$\frac{(\log k)^2}{108} \leq D_{92}(k) \leq \lceil (\log k)^2 \rceil + 1.$$

Thus the central dimension-order problem has the **complete answer** $\Theta((\log k)^2)$ **for all** k . The proof first resolves dyadic $k = 2^r$ by linear codes and then glues the binary expansion with a one-dimensional Gaussian selector.

Let $F_d(k)$ denote the global optimum of

$$P(\mathcal{A}) = \sum_{u=1}^k \left\| \int_{A_u} x d\gamma_d(x) \right\|^2$$

over arbitrary measurable partitions of standard Gaussian $\mathbb{R}^d$ into exactly k regions of mass $1/k$. Let $F_\infty(k) = F_{k-1}(k)$.

Theorem 1 (dyadic component of the all-integer theorem). There is an absolute integer r_0 such that for every integer $r \geq r_0$, writing $k = 2^r$ and $L = \log k$, there exists an exact equiprobable k -cell Gaussian partition in

$$d = \lceil L^2 \rceil$$

dimensions with

$$F_d(k) \geq F_\infty(k) - \frac{100}{k}. \tag{1}$$

The partition is obtained by taking the largest of k explicitly specified unit Gaussian linear scores indexed by $\mathbb{F}_2^r$.

Together with the universal spherical-cap lower bound, the minimum dimension $D_{100}(2^r)$ required to achieve error at most $100/2^r$ satisfies

$$\boxed{D_{100}(2^r) = \Theta(r^2) = \Theta((\log k)^2).} \quad (2)$$

Theorem 11 below removes the dyadic restriction with a one-coordinate exact Gaussian selector. Neither result identifies the finite- k global optimizer.

The construction is probabilistic in the proof of existence. After choosing a successful binary matrix, all its entries are ordinary finite integers and the induced Gaussian cells are completely deterministic. Exact equal masses follow from an orthogonal group action; there is no probabilistic mass correction.

1. Linear-code orbit and pairwise independence

Let $k = 2^r$, $m = \lceil (\log k)^2 \rceil$. Choose column vectors $g_1, \dots, g_m \in \mathbb{F}_2^r$ independently and uniformly. For each $u \in \mathbb{F}_2^r$, set

$$v_u = m^{-1/2}((-1)^{u \cdot g_1}, \dots, (-1)^{u \cdot g_m}) \in S^{m-1}. \quad (3)$$

Here the inner products inside the exponents are taken in $\mathbb{F}_2$.

Lemma 2 (exact transitivity when the matrix has full rank). If $g_1, \dots, g_m$ span $\mathbb{F}_2^r$, the v_u are pairwise distinct. The Gaussian score-maximizing sets

$$A_u = \{x : \langle v_u, x \rangle > \langle v_w, x \rangle \text{ for all } w \neq u\} \quad (4)$$

partition $\mathbb{R}^m$ modulo null hyperplanes, and satisfy $\gamma_m(A_u) = 1/k$ exactly. Moreover,

$$\boxed{P(\mathcal{A}) \geq \frac{1}{k} \left(\mathbb{E} \max_u \langle v_u, G \rangle \right)^2.} \quad (5)$$

Proof. Distinctness follows since $v_u = v_w$ would imply $(u - w) \cdot g_j = 0$ for every j , contradicting the span condition. For each $a \in \mathbb{F}_2^r$, the diagonal orthogonal transformation

$$D_a = \text{diag}((-1)^{a \cdot g_1}, \dots, (-1)^{a \cdot g_m})$$

satisfies $D_a v_u = v_{u+a}$. It carries each score-max cell to the corresponding translated cell. Gaussian rotational invariance makes all cell masses equal; they sum to one.

Writing $b_u = \int_{A_u} x d\gamma_m(x)$, we have

$$\mathbb{E} \max_u \langle v_u, G \rangle = \sum_u \langle v_u, b_u \rangle \leq \sqrt{k} \left(\sum_u \|b_u\|^2 \right)^{1/2}.$$

Squaring gives (5). $\square$

Let the score for message u be $X_u = \langle v_u, G \rangle$, where $G_1, \dots, G_m$ are independent $N(0, 1)$. Condition on the entire Gaussian vector G , and put $a_j = G_j / \sqrt{m}$. The score for $u = 0$ is the deterministic (given G) value

$$X_0 = \sum_j a_j. \quad (6)$$

Lemma 3 (pairwise-independent conditional exceedances). Over the random binary columns, for every nonzero u , the conditional distribution of X_u is that of the Rademacher-weighted sum $S_a = \sum_j a_j \epsilon_j$, with independent symmetric signs ϵ_j . For every two distinct nonzero u, w , the conditional random variables X_u, X_w are independent. Consequently, for any real threshold t , if $q_a(t) = \Pr_\epsilon(S_a \geq t) > 0$, then

$$\boxed{\Pr_g \{ \max_u X_u < t \mid G \} \leq \frac{1}{(k-1)q_a(t)}.} \quad (7)$$

Proof. The $\mathbb{F}_2$ -linear functional $g \mapsto u \cdot g$ is unbiased for nonzero u . Distinct nonzero u, w are linearly independent over $\mathbb{F}_2$, so $g \mapsto (u \cdot g, w \cdot g)$ is uniform on $\mathbb{F}_2^2$. Independent columns yield independent sign vectors for the two messages, conditional on fixed G .

Let $N_t = \#\{u \neq 0 : X_u \geq t\}$. Conditional pairwise independence gives $\mathbb{E}_g N_t = (k-1)q_a(t)$ and $\text{Var}_g N_t \leq (k-1)q_a(t)$. Chebyshev's inequality for the event $N_t = 0$ proves (7). $\square$

Thus the code structure supplies exact mass symmetry and, independently, a second-moment method. No bound on the worst correlation of two codewords is required.

2. Tilted Rademacher anti-concentration

Write

$$Q_2 = \sum_j a_j^2, \quad Q_4 = \sum_j a_j^4, \quad G = \{1/2 \leq Q_2 \leq 2, \quad Q_4 \leq 10/m\}. \quad (8)$$

Lemma 4 (typical Gaussian weights). For $G_j \stackrel{\text{iid}}{\sim} N(0, 1)$,

$$\boxed{\Pr_G(G^c) \leq 10/m.} \quad (9)$$

Proof. Direct normal moment calculation gives

$$\mathbb{E}Q_2 = 1, \quad \text{Var } Q_2 = 2/m, \quad \mathbb{E}Q_4 = 3/m, \quad \text{Var } Q_4 = 96/m^3,$$

since $\mathbb{E}G^4 = 3$ and $\mathbb{E}G^8 = 105$. Chebyshev bounds the first failure by $8/m$ and the second by $96/(49m) < 2/m$. Union bounding proves (9). $\square$

For nonzero weights a and $\lambda > 0$, introduce the exponentially tilted sign law

$$\Pr_\lambda(\epsilon_j = e) = \frac{\exp(\lambda a_j e)}{2 \cosh(\lambda a_j)}, \quad e = \pm 1.$$

Set

$$\Lambda(\lambda) = \sum_j \log \cosh(\lambda a_j), \quad \mu = \Lambda'(\lambda) = \sum_j a_j \tanh(\lambda a_j), \quad V = \Lambda''(\lambda) = \sum_j a_j^2 \text{sech}^2(\lambda a_j). \quad (10)$$

Classical Berry–Esseen input. For independent centered real random variables W_j with total variance $V > 0$, the inequality

$$\sup_x \left| \Pr\left(\frac{\sum W_j}{\sqrt{V}} \leq x\right) - \Phi(x) \right| \leq \frac{\sum_j \mathbb{E}|W_j|^3}{V^{3/2}} \quad (11)$$

holds (the deliberately nonoptimal constant 1 is sufficient). For a published strictly stronger **non-identically distributed** version with constant $0.5591 < 1$, see I. S. Tyurin, *A Refinement of the Remainder in the Lyapunov Theorem, Theory of Probability & Its Applications* **56** (2012), 693–696, DOI 10.1137/S0040585X9798572X. We apply (11) only to bounded independent centered tilted signs and explicitly bound the numerator and denominator below. This standard result is the sole non-elementary probability theorem imported into the proof.

Lemma 5 (uniform moderate-deviation lower tail). There exists an absolute L_0 (one may take $L_0 = 10^{10}$) such that whenever $L = \log k \geq L_0$, $m = \lceil L^2 \rceil$, and $G \in G$, put

$$\lambda = \sqrt{\frac{2L}{Q_2}}, \quad B_0 = \frac{1}{2} \log L + 12, \quad s_y = \mu - \frac{B_0 + y}{\lambda} \quad (0 \leq y \leq \lambda/4). \quad (12)$$

Then

$$\boxed{\Pr_\epsilon\{S_a \geq s_y\} \geq \frac{e^{-L+B_0+y-1}}{20\lambda}.} \quad (13)$$

Consequently the randomly generated code satisfies

$$\boxed{\Pr\{\max_u X_u < s_y \mid G\} \leq \eta e^{-y}, \quad \eta = 80e^{-11} < 1/10.} \quad (14)$$

Proof. On G , $\sqrt{L} \leq \lambda \leq 2\sqrt{L}$. Since $\text{sech}^2 x \geq 1 - x^2$,

$$V \geq Q_2 - \lambda^2 Q_4 \geq 1/2 - 40L/m \geq 1/4$$

for $L \geq 160$, while $V \leq Q_2 \leq 2$. The centered tilted variables are bounded in absolute value by $2|a_j|$. Hence

$$\sum_j \mathbb{E}_\lambda |W_j|^3 \leq 8 \sum_j |a_j|^3 \leq 8\sqrt{Q_2 Q_4} \leq 8\sqrt{20/m}.$$

By (11), the error for a tilted CDF is at most $64\sqrt{20/m} < 288/\sqrt{m}$.

For $L \geq 10^{10}$ we have $B_0 \leq \lambda/4$. Therefore for $0 \leq y \leq \lambda/4$, the interval

$$I_y = [s_y, s_y + 1/\lambda]$$

lies, after centering at μ and dividing by $\sqrt{V}$, inside $[-1, 0]$. Its standardized length is at least $1/(2\lambda)$. A standard normal has density at least $\varphi(1) > 1/5$ on this range. Thus its probability in I_y is at least $1/(10\lambda)$. Two applications of (11) cost less than $576/\sqrt{m} \leq 1/(20\lambda)$ for $L \geq 10^{10}$. Consequently

$$\Pr_\lambda(S_a \in I_y) \geq 1/(20\lambda). \quad (15)$$

The exact change-of-measure identity gives

$$\Pr_\epsilon(S_a \geq s_y) \geq \frac{1}{20\lambda} \exp\{\Lambda(\lambda) - \lambda(s_y + 1/\lambda)\}. \quad (16)$$

For every real x ,

$$\log \cosh x - x \tanh x + x^2/2 \geq 0,$$

because the left side is even, vanishes at zero, and its derivative for $x > 0$ is $x \tanh^2 x \geq 0$. Summing over $x = \lambda a_j$, we obtain

$$\Lambda(\lambda) - \lambda\mu \geq -\lambda^2 Q_2/2 = -L.$$

Inserting s_y in (16) proves (13).

Finally, using $\lambda \leq 2\sqrt{L}$, $k - 1 \geq k/2$, and $k = e^L$,

$$(k - 1)q_a(s_y) \geq \frac{e^{B_0 + y - 1}}{40\lambda} \geq \frac{e^{11 + y}}{80}.$$

Apply Lemma 3 to obtain (14). The inequality $\eta < 1/10$ follows from $e > 2$, hence $e^{11} > 2^{11} > 800$. $\square$

3. From exceedance tails to a sharp expected maximum

The next lemma converts the preceding exponentially improving threshold estimate into an **expectation bound at the $1/\sqrt{L}$ scale**. This is the crucial step that a single constant-probability exceedance estimate would *not* provide.

Lemma 6 (conditional expected-score estimate). For $L \geq 10^{10}$, $m = \lceil L^2 \rceil$, and every Gaussian vector $G \in G$, the random generator satisfies

$$\boxed{\mathbb{E}_g[M \mid G] \geq \mu - \frac{B_0}{\lambda} - \frac{\eta}{\lambda} - \eta(4\sqrt{L} + |X_0|)e^{-\lambda/4}, \quad M = \max_{u \in \mathbb{F}_2^r} X_u.} \quad (17)$$

Proof. As y runs from 0 to $y_* = \lambda/4$, the threshold s_y decreases linearly with slope $-1/\lambda$. The pointwise identity

$$(s_0 - M)_+ = \frac{1}{\lambda} \int_0^{y_*} \mathbf{1}_{\{M < s_y\}} dy + (s_{y_*} - M)_+$$

holds, regardless of any ties. Using (14), the expected integral is at most η/λ . Since $M \geq X_0$, and $s_{y_*} \leq \mu \leq \lambda Q_2 \leq 4\sqrt{L}$,

$$(s_{y_*} - M)_+ \leq (4\sqrt{L} + |X_0|) \mathbf{1}_{\{M < s_{y_*}\}}.$$

Its expectation is at most $\eta(4\sqrt{L} + |X_0|)e^{-\lambda/4}$. Finally $M \geq s_0 - (s_0 - M)_+$. Taking expectations proves (17). $\square$

Lemma 7 (mean over generators and Gaussian data). For $L \geq 10^{10}$,

$$\boxed{\mathbb{E}_{g,G} M \geq \sqrt{2L} - \frac{\log L}{2\sqrt{2L}} - \frac{27}{\sqrt{L}}.} \quad (18)$$

Proof. Let $\sigma = \sqrt{Q_2}$, so $\lambda = \sqrt{2L}/\sigma$. The elementary inequality

$$\tanh x \geq x - x^3/3 \quad (x \geq 0)$$

follows on differentiation from $x^2 - \tanh^2 x \geq 0$. It implies

$$\mu \geq \lambda Q_2 - \lambda^3 Q_4/3.$$

On G , $\sigma \geq 1/\sqrt{2}$, so

$$\mu \geq \sqrt{2L} \sigma - \frac{8}{3} L^{3/2} Q_4.$$

Since $\mathbb{E} Q_4 = 3/m$,

$$\mathbb{E}[\mu \mathbf{1}_G] \geq \sqrt{2L} \mathbb{E}[\sigma \mathbf{1}_G] - 8L^{3/2}/m. \quad (19)$$

Because $|\sqrt{x} - 1| \leq |x - 1|$ for $x \geq 0$, $\mathbb{E}|\sigma - 1| \leq \sqrt{\text{Var } Q_2} = \sqrt{2/m}$. Cauchy-Schwarz, $\mathbb{E}\sigma^2 = 1$, and Lemma 4 give

$$\mathbb{E}[\sigma \mathbf{1}_G] \geq 1 - \sqrt{2/m} - \sqrt{10/m}.$$

Using $m \geq L^2$,

$$\mathbb{E}[\mu \mathbf{1}_G] \geq \sqrt{2L} - 7/\sqrt{L} - 8/\sqrt{L}. \quad (20)$$

Also, $\mathbb{E}\sigma \leq 1$ by Jensen's inequality, so

$$\mathbb{E}[(B_0/\lambda) \mathbf{1}_G] \leq \frac{B_0}{\sqrt{2L}} = \frac{\log L}{2\sqrt{2L}} + \frac{12}{\sqrt{2L}}.$$

The last constant term is less than $9/\sqrt{L}$. Furthermore $\eta/\lambda \leq 1/\sqrt{L}$ on G .

For $L \geq 10^{10}$, the elementary inequality $e^x \geq x^4/4!$ and $\lambda \geq \sqrt{L}$ give

$$\eta e^{-\lambda/4} (4\sqrt{L} + \mathbb{E}|X_0|) \leq 1/\sqrt{L}.$$

Indeed $X_0 \sim N(0, 1)$, $\mathbb{E}|X_0| \leq 1$, and $e^{-\sqrt{L}/4} \leq 6144/L^2$.

Finally, on G^c we still have $M \geq X_0$. Since $\mathbb{E}X_0^2 = 1$, Lemma 4 yields

$$\mathbb{E}[M \mathbf{1}_{G^c}] \geq -\sqrt{\text{Pr}(G^c)} \geq -\sqrt{10/m} \geq -1/\sqrt{L}$$

for $L \geq 10^{10}$. Combining all the displayed bounds with Lemma 6 gives (18): the six separate $1/\sqrt{L}$ losses have coefficients at most 7, 8, 9, 1, 1, 1, totaling 27. $\square$

4. Enforcing full rank without losing the estimate

The previous expectation averages over all binary generators, including the rare matrices whose columns fail to span the message space. We now remove them without appealing to numerical rank search.

Lemma 8 (full-rank extraction). For $L \geq 10^{10}$, there is a binary $r \times m$ matrix of rank exactly r whose codeword Gaussian maximum satisfies

$$\mathbb{E}_G \max_u \langle v_u, G \rangle \geq \sqrt{2L} - \frac{\log L}{2\sqrt{2L}} - \frac{30}{\sqrt{L}}. \quad (21)$$

Proof. For any nonzero $u \in \mathbb{F}_2^r$, the probability that $u \cdot g_j = 0$ for every j is 2^{-m} . The rank-failure event is the union of these events, hence

$$p_{\text{bad}} \leq (k-1)2^{-m} \leq \exp(L - m \log 2) \leq L^{-2}$$

for $L \geq 10^{10}$, since $\log 2 > 2/3$ and $m \geq L^2$.

For every fixed generator, each score X_u is a standard normal. A Gaussian union bound yields

$$\Pr_G(M \geq t) \leq ke^{-t^2/2} \quad (t \geq 0).$$

Moreover $M \geq X_0$, so the negative part satisfies $M_- \leq (X_0)_-$. Integrating the positive tail at the split $\sqrt{2L}$, we get

$$\mathbb{E}_{g,G} M_+^2 \leq 2L + 2, \quad \mathbb{E}_{g,G} M_-^2 \leq 1.$$

Therefore

$$\mathbb{E}_{g,G} M^2 \leq 2L + 3.$$

By Cauchy–Schwarz,

$$|\mathbb{E}[M \mathbf{1}_{\{\text{rank} < r\}}]| \leq \sqrt{(2L+3)p_{\text{bad}}} \leq \frac{2}{\sqrt{L}}.$$

Lemma 7 now shows that the unconditional average of M over **full-rank** matrices, normalized by their probability, is at least the right side of (21); the normalization can only increase a positive lower bound for all sufficiently large L . At least one full-rank matrix attains this average or better. $\square$

5. Proof of the main theorem and optimal order

Proof of Theorem 1. Choose the full-rank matrix of Lemma 8. Its score-max partition has exactly equal masses by Lemma 2, and

$$kP(\mathcal{A}) \geq \left(\sqrt{2L} - \frac{\log L}{2\sqrt{2L}} - \frac{30}{\sqrt{L}} \right)^2.$$

For $L \geq 10^{10}$ the bracket is positive. Expanding the square and discarding nonnegative terms gives

$$kP(\mathcal{A}) \geq 2L - \log L - 60\sqrt{2} \geq 2L - \log L - 85,$$

since $60\sqrt{2} < 85$.

The scalar quantile ceiling

$$h(1/k)^2 \leq 2L - \log L + 3$$

follows directly from Mills' inequalities. Indeed, let $t = \Phi^{-1}(1 - 1/k) > 1$. Then $h(1/k) \leq t + 1/t$, so $h(1/k)^2 \leq t^2 + 3$; and $L \geq t^2/2 + \log t + \frac{1}{2} \log(2\pi)$. If $t^2 \geq L$, this forces $t^2 \leq 2L - \log L$. Otherwise $t^2 < L \leq 2L - \log L$. Thus the ceiling holds in both cases. The individually optimal halfspace upper envelope is $U_k = h(1/k)^2/k$; thus $U_k - P(\mathcal{A}) \leq 88/k$. Because $F_\infty(k) \leq U_k$,

$$P(\mathcal{A}) \geq F_\infty(k) - 88/k \geq F_\infty(k) - 100/k,$$

proving (1).

For the optimal dimension order, the independently established spherical-cap converse says that any partition with score at least $F_\infty(k) - 100/k$ must, for all sufficiently large k , have

$$d \geq \frac{(\log k)^2}{116}.$$

Our construction has $d = \lceil (\log k)^2 \rceil$. Restricting to the dyadic subsequence $k = 2^r$ proves (2). $\square$

6. Binary-block completion for all cell counts

The dyadic theorem already determines the optimal dimensional **order for all sufficiently large integers k** , using a nonuniform one-dimensional *selector* and a bounded-entropy binary expansion. This section removes the dyadic restriction.

Let r_0 be the fixed exponent threshold in Theorem 1, and set $Q_0 = 2^{r_0}$.

Lemma 9 (sharp binary-expansion entropy). Write any positive integer as a sum of distinct powers of two, ordered strictly decreasing:

$$k = q_1 + \cdots + q_s, \quad q_j = 2^{r_j}, \quad r_1 > \cdots > r_s \geq 0,$$

and put $w_j = q_j/k$. Then

$$H(w) := \sum_{j=1}^s w_j \log(1/w_j) < 2 \log 2 \quad (s < \infty), \quad (22)$$

with equality approached as $k = 2^s - 1 \rightarrow \infty$, so **the coefficient $2 \log 2$ is best possible**. Furthermore,

$$\sum_{j: q_j < Q_0} q_j < Q_0. \quad (23)$$

Proof. Since q_j is a power of two and every subsequent q_ℓ is a distinct smaller power,

$$\sum_{\ell > j} q_\ell \leq q_j - 1 < q_j.$$

Put $S_j = \sum_{\ell \geq j} w_\ell$, $S_{s+1} = 0$. Then $S_{j+1} < w_j$, hence

$$\frac{S_{j+1}}{S_j} < \frac{1}{2}, \quad S_j \leq 2^{1-j}.$$

Define $\alpha_j = w_j/S_j$ and the binary entropy function

$$h_2(t) = -t \log t - (1-t) \log(1-t), \quad 0 \log 0 := 0.$$

The exact entropy chain identity is

$$H(w) = \sum_{j=1}^s S_j h_2(\alpha_j).$$

Indeed expansion of $S_j h_2(\alpha_j)$ gives $-w_j \log w_j - S_{j+1} \log S_{j+1} + S_j \log S_j$, and the last two terms telescope. Because $h_2(t) \leq \log 2$ for every $t \in [0, 1]$,

$$H(w) \leq (\log 2) \sum_{j=1}^s S_j \leq (\log 2) \sum_{j=1}^s 2^{1-j} < 2 \log 2.$$

To see optimality, put $k = 2^s - 1$, so $w_j = 2^{s-j}/(2^s - 1)$. For each fixed j , $w_j \rightarrow 2^{-j}$, and the geometric bound above permits dominated convergence in the entropy series. The limiting geometric distribution has entropy $\sum_{j \geq 1} 2^{-j} \log 2 = 2 \log 2$. Thus the coefficient cannot be improved.

Finally, the powers of two smaller than $Q_0 = 2^{r_0}$ have total sum at most $1 + 2 + \dots + 2^{r_0-1} = Q_0 - 1$, proving (23). $\square$

Lemma 10 (one Gaussian coordinate glues exact mass blocks). Suppose $k = \sum_j q_j$ with positive integers q_j , and for each j there is a Gaussian partition $\mathcal{B}^{(j)}$ of a common $\mathbb{R}^m$ into exactly q_j cells of measure $1/q_j$, with squared-first-moment objective P_j . There is then an exactly equiprobable k -cell partition of $\mathbb{R}^{m+1}$ with objective

$$P_{\text{glue}} \geq \frac{1}{k^2} \sum_j q_j^2 P_j. \quad (24)$$

Proof. Take an independent standard Gaussian selector $T \in \mathbb{R}$ and partition its real line into Borel intervals E_j of probabilities $\Pr(T \in E_j) = q_j/k$. Such intervals exist uniquely using the continuous Gaussian CDF.

For every local cell $B_u^{(j)}$, define the final cell $A_{j,u} = E_j \times B_u^{(j)}$. Independence gives

$$\gamma_{m+1}(A_{j,u}) = (q_j/k)(1/q_j) = 1/k.$$

The first moment in the m data coordinates equals

$$\int_{A_{j,u}} x_{\text{data}} d\gamma_{m+1} = \frac{q_j}{k} \int_{B_u^{(j)}} x d\gamma_m.$$

Summing squared norms over all final cells yields exactly $\frac{1}{k^2} \sum_j q_j^2 P_j$ from the data coordinates. Each cell's selector-coordinate moment contributes a further nonnegative square. This proves (24). $\square$

Theorem 11 (universal quadratic-logarithmic dimension, all integers). With $Q_0 = 2^{r_0}$ as above, take the explicit threshold $K_0 = Q_0^2$. Then **for every** integer $k \geq K_0$, writing $L = \log k$, there is a Gaussian partition of $\mathbb{R}^d$ into exactly k cells, each of measure $1/k$, with

$$d \leq \lceil L^2 \rceil + 1 \leq L^2 + 2, \quad P(\mathcal{A}) \geq F_\infty(k) - \frac{95}{k}. \quad (25)$$

More precisely, its objective meets the universal one-cell halfspace envelope:

$$U_k - P(\mathcal{A}) \leq 92/k. \quad (26)$$

Let $D_{92}(k)$ be the smallest Gaussian ambient dimension allowing an exactly equiprobable k -cell partition with objective at least $F_\infty(k) - 92/k$. Combining (25) with the spherical-cap dimension converse gives, for all sufficiently large integers k ,

$$\frac{(\log k)^2}{108} \leq D_{92}(k) \leq \lceil (\log k)^2 \rceil + 1. \quad (27)$$

Consequently,

$$D_{92}(k) = \Theta((\log k)^2) \quad (k \rightarrow \infty) \text{ for all integers } k. \quad (28)$$

This settles the **optimal order** of dimensional growth for bounded additive $O(1/k)$ approximation of the global equal-mass Gaussian first-moment optimum.

Proof. Fix k and write its binary expansion as in Lemma 9. Take a common data Gaussian space of dimension $m = \lceil L^2 \rceil$.

For every binary block $q_j \geq Q_0$, Theorem 1's **proved construction** has local dimension $\lceil (\log q_j)^2 \rceil \leq m$. After padding with unused independent Gaussian coordinates, it defines a q_j -cell equiprobable partition in the common space satisfying the stronger *halfspace* estimate

$$P_j \geq \frac{h(1/q_j)^2 - 88}{q_j}. \quad (29)$$

Indeed the proof of Theorem 1 already obtained the stronger estimate with 88 as displayed.

For a smaller block $q_j < Q_0$, choose any equiprobable Gaussian q_j -partition of the same data space, e.g. the consecutive quantile intervals of its first coordinate. Only the trivial bound $P_j \geq 0$ is used for these blocks.

Apply Lemma 10. Let W_{sm} be the sum of $w_j = q_j/k$ over small blocks. Then

$$\begin{aligned} kP_{\text{glue}} &\geq \sum_{j: q_j \geq Q_0} w_j (h(1/q_j)^2 - 88), \\ h(1/k)^2 - kP_{\text{glue}} &\leq 88 + \sum_{j: q_j \geq Q_0} w_j [h(1/k)^2 - h(1/q_j)^2] + W_{\text{sm}} h(1/k)^2. \end{aligned}$$

Define $h(1) = 0$, the continuous extension of the upper-tail conditional mean to the sure event. The squared Gaussian upper-tail conditional mean is globally 2-Lipschitz in log tail mass: for $q_j \leq k$,

$$0 \leq h(1/k)^2 - h(1/q_j)^2 \leq 2 \log(k/q_j) = 2 \log(1/w_j).$$

By Lemma 9, the total difference term is at most $2H(w) < 4 \log 2$. Also $W_{\text{sm}} < Q_0/k$ and the elementary Gaussian exponential-moment bound gives $h(1/k)^2 \leq 2L$. Hence

$$h(1/k)^2 - kP_{\text{glue}} \leq 88 + 4 \log 2 + \frac{2Q_0 L}{k}. \quad (30)$$

For $k \geq K_0 = Q_0^2$, the function $x \mapsto \log x/x$ is decreasing on $x \geq e$, so

$$\frac{2Q_0 \log k}{k} \leq \frac{4 \log Q_0}{Q_0} < 1.$$

The last inequality holds for $Q_0 \geq 2^{10}$ by monotonicity, since $4 \log(2^{10})/2^{10} < 40/1024 < 1$. Our r_0 is much larger than ten. Since $4 \log 2 < 3$ (as $e^{3/4} > 1 + 3/4 + (3/4)^2/2 > 2$), the right side of (30) is strictly below $88 + 3 + 1 = 92$. This proves (26). Because $F_\infty(k) \leq U_k$, (25) follows.

For the matching lower dimension, the independently proved spherical-cap converse says that for each fixed additive tolerance C , every sufficiently accurate partition must have dimension at least $L^2/(C + 16)$. Put $C = 92$ to obtain (27). The sandwich proves (28). $\square$

What is now resolved, and what remains

The order $\Theta((\log k)^2)$ in (28) is a **complete all-integer dimension-rate classification at the level of orders** for a fixed bounded $1/k$ additive error. The proof explicitly handles arbitrarily many binary digits and includes extreme binary-expansion boundary cases.

The theorem does not determine the smallest possible dimensional *constant*, the best additive-error constant, the exact finite- k optimizer, or efficient algorithms for the binary generator matrices. Those remain valid further research targets.

7. Source status and remaining global problems

Theorems 1 and 11 establish a **matching-order quadratic-logarithmic dimensional law** for all sufficiently large cell counts, not only dyadic subsequences. The lower and upper constants $1/108$ and 1 , and the additive error $92/k$, are not claimed sharp.

The only substantial external probabilistic input is the classical Berry–Esseen inequality for independent, non-identically distributed summands. The Gaussian and finite-field calculations are derived in detail here. No numerical solver, simulation or optimizer assertion is used as a theorem.

The matrix construction is by the probabilistic method: an admissible finite full-rank binary matrix is proved to exist for every sufficiently large dyadic block, but efficiently computing it is not established. The Gaussian selector and binary expansion are explicit deterministic operations once these matrices are fixed.

The **core dimension-order problem for fixed bounded additive error is resolved at order level**. Remaining open questions include the sharp leading dimension coefficient, best additive constant, effective polynomial-time construction, stability and equality cases, and the exact fixed- k Standard Simplex optimality question. No historical world-first claim is made pending full specialist literature comparison and external peer review.

Proof-interface checks, attribution and dependency limitations are recorded in `AUDIT.md` and `LITERATURE.md`.